

ML Model: Rented Bikes Demand in Seoul

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Abstract—The objective of this project is to develop a machine learning model. The problem adjacent to this objective is the prediction of rented bikes in Seoul based on published data donated to the UC Irvine Machine Learning Repository. It contains 8760 instances, chronologically ordered by hour, from 1 December 2017 to 30 November 2018. In addition to its data limitations, the resulting model should be able to predict the demand based on measurable features. The possibility of feeding up-to-date data in a larger sample could be resourceful in the prediction of demand. This paper is a justification for the process from the raw data up to the final model and a comparison of versions. The paper will explore all decisions made and the reasoning behind them and their justification, as well as the analysis of all the results. The procedure of this project starts with an ETL method for raw data treatment, then an EDA for a better understanding of the situation, and a PCA for final consideration in data treatment. In the development of the model, it will focus on two approaches: a framework and non-framework code, then comparing the results.

Index Terms—Machine Learning, ETL, EDA, PCA, Seoul Bike Sharing Demand, Prediction Model

I. INTRODUCTION

It's possible to make a prediction by pure statistics; however, the first results on this project look forward to taking advantage of machine learning and producing an accurate prediction. Although the development and results for this project are limited and confined by the sandbox of raw data, it should be a first step in understanding an introduction to machine learning for me and the reader, an exercise of the mind so better knowledge can be produced. And if someday it were to create an absolute model that could predict the demand with such precision for bikes, transportation in Seoul or other cities, this and all people working on the same data and similar projects would be the foundation and first step for it.

II. EXTRACT, TRANSFORM, LOAD (ETL)

A. Data Extraction, Initial Inspection and Understanding

To produce an ETL we need to understand the raw dataset. The dataset comprises 14 features and 8760 instances in particular. We are talking about 365 days from 1 December 2017 to 30 November 2018, divided by hour instances; besides, out of those 14 features, the goal of the project is to predict the feature Rented Bike Count given the other parameters.

Furthermore, for these variables is important to understand that: In the feature Visibility, the method of measurement or instrument is unknown, but every unit is expressed as "10m" referring to a longitudinal distance in which it is visible, probably due to weather factors. The Dew point is

TABLE I
RAW DATASET DESCRIPTION

Feature	Description	Type
Date	Format dd/mm/yyyy	Object
Rented Bike Count	Total bikes rented per hour (Target variable)	Integer
Hour	Hour of the day (0–23)	Integer
Temperature	Ambient temperature (°C)	Float
Humidity	Relative humidity (%)	Integer
Wind speed	Wind speed (m/s)	Float
Visibility	Visibility distance in 10m units	Integer
Dew point temperature	Temperature at which air becomes fully saturated (°C)	Float
Solar Radiation	Accumulated solar energy (MJ/m^2)	Float
Rainfall	Rainfall amount (mm)	Float
Snowfall	Snowfall amount (cm)	Float
Seasons	Seasonal periods of the year	Object
Holiday	Holiday indicator (Holiday / No Holiday)	Object
Functioning Day	Operational status of the rental system	Object

the temperature at which air becomes fully saturated with water vapor and cannot hold moisture any longer. Is important when predicting weather forecasts due to its relation with other factors. Solar Radiation is measured in MegaJoules per square meter and represents the total accumulated energy produced by the sun in a lapse of time (Hour in this case). The Date its resourceful to understand results and the chronology of events; however, it won't be fed to the model since it isn't helpful for a prediction, just for a search. Worth mentioning that it had no null values for any feature.

B. Understandable Data for the Model

The first step is the transformation of all Objects and Strings into understandable data. For Holiday, we transformed it into a binary integer value; it either is a holiday or isn't. Seasons were transformed into an integer variable, dummy 1-4 starting with Winter. Functioning day was also transformed into a binary integer value

C. Visibility Uncertainty Translation

For the visibility uncertainty on how it was measured, it was translated into a ratio of how much was visible for users when riding a bike, where 0 is no visibility and 1 is the maximum visibility; this was for better digestion for our model. This could be done because, after data inspection, it had a maximum

value of 2000 and referred to a distance value, so it could be scaled.

D. Functionality Imputation

For the visibility uncertainty on how it was measured, it was translated into a ratio of how much was visible for users when riding a bike, where 0 is no visibility and 1 is the maximum visibility; this was for better digestion for our model. This could be done because, after data inspection, it had a maximum value of 2000 and referred to a distance value, so it could be scaled.

E. Date Feature Compensation

As the date won't be fed to compensate the feature, we can extract a binary value that can help our model better understand the routines. From the date, using pandas pd.to_datetime, we can get an int binary value for the weekend feature.

F. ETL Dataset

Up to this point, we do not possess null instances across all the data. The data has been transformed into digestible, understandable units and values. All calculations for the new data were successful and do not contain errors. The data was treated to the optimal type. Finally, all instances are still in chronological order. We end with this post-ETL dataset.

TABLE II
POST ETL DATASET

Feature	Type	Feature Class
Date	Value	Map
Rented Bike Count	Integer	Y
Hour	Integer	X
Temperature	Float	X
Humidity	Integer	X
Wind speed	Float	X
Visibility	Float,Ratio	X
Dew point temperature	Float	X
Solar Radiation	Float	X
Rainfall	Float	X
Snowfall	Float	X
Seasons	Int	X
Holiday	Int,Binary	X
Functioning Day	Int,Binary	MAP
Weekend	Int,Binary	X

G. Division of classes

We can now produce our division of classes for the model. The division percentage is set to be 90% of the data destined to train and the remaining 10% to test the model. To split the data, we have to consider the factor known as seasonality or trend. In finance and economics is best noticed by the annual seasons in which data patterns emerge and behave differently between seasons. Therefore, it was decided that the division of data will be divided equally from all seasons. 36 days, representing approximately 10% of the total data, will be taken from the last days of each season, meaning that for each season

the last 9 days, or 216 instances, will be destined for testing. The Y train and test subset will contain only Rented Bike Count; the X subset will have the rest, excluding the Date and Functioning Day. Recycling these features of data we create also a subset defined as Map that will contain these reused excluded features, as well as all the instances in order with the corresponding season, the subset it belongs to (train or test), and a local ID for the subset. This is with the purpose of having a map that will act as a tool when processing results, to graph all data in chronological order and explain possible anomalies due to the imputation or division of continuous data. The justification for creating the map is that it provides the information to order and filter data. After the division, we are left with the next subsets of data.

TABLE III
REPRESENTATION OF DIVISION OF DATA

Class/Subset	Description
X test	Last X features from the last 216 instances from each season
Y test	Last Y features from the last 216 instances from each season
X train	Left X features from the split from each season
Y train	Left Y features from the split from each season
Map	All map features sorted by an index, keeping chronological order

III. EXPLORATORY DATA ANALYSIS (EDA)

The EDA of interest only falls under the training features, including the independent variable. At this point, we can produce copies of our subset to model at our will without losing the information for the model.

A. Distribution with Histogram

For the analysis of histograms, we discriminate the data for Hours, Seasons, and Weekend, as that categorical data frequency won't add information. We then can produce of all histograms for all continuous data, including the histogram for Holiday. We are left with the following graphs.

1.1 The distribution for the Rented Bike Count is shown as a positively skewed distribution with a long tail. Most of the data falls below 500; we can assume that most instances or hours present fewer than 500 rentals.

1.2 The distribution of Temperature seems a bit symmetrical, with a higher frequency at approximately 20 degrees. Most instances during the year of data collected were about 20 degrees Celsius in Seoul.

1.3 The distribution of humidity, although it almost looks symmetrical it has a high bias towards high values, with only a few occurrences around 0% and a high peak at 100%. With peaks at 50 and 100, we can assume that most instances presented at least a slightly higher humidity percentage in the data collected in Seoul.

1.4 The wind speed distribution presents a positively skewed distribution with a long tail distribution with most values

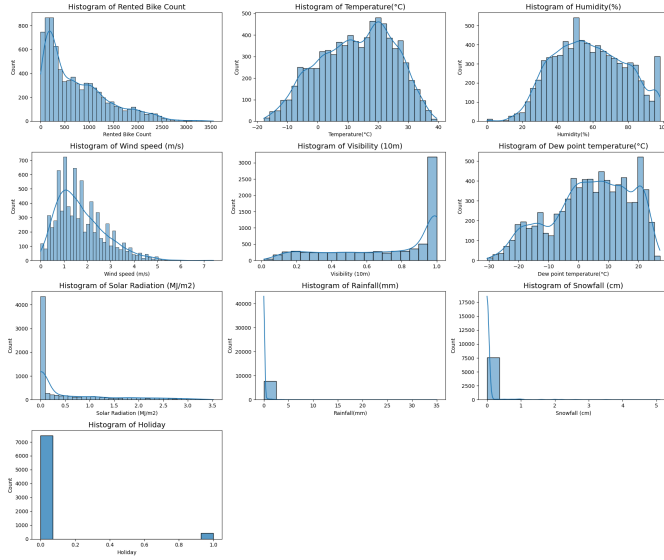


Fig. 1. Histograms of continuous variables and holiday distribution.

falling around 1m/s. The majority of instances presented less than 3m/s wind speed readings in Seoul.

1.5 The distribution of visibility is highly negatively skewed, with the most frequent value being 1 also understood that the majority of instances present 100% or complete of visibility.

1.6 The distribution of Dew temperature seems to present a non-symmetrical distribution, with no high bias for the left side. It presents two pinnacles: first at around -20 degrees, then falls into a valley around -10 and rises to a higher pinnacle and larger group of frequency around 10 degrees, with a noticeable single peak at 20 degrees. We can assume that it presents a bimodal characteristic and that two groups of data could form around -20 degrees and 10 degrees of dew point; the dew point is ultimately noticed by people as it has a highly physical relation with weather. Therefore, there could be the possibility that it could provide a high numerical relationship when comparing the output with 2 possible classes. It could also be useful that the 20 degree peak has a strong numerical relationship with Temperature.

1.7 The histogram of solar radiation presents a positveily skew frequency, with most of the data falling near 0 Mj/m2; most instances in Seoul present around 0 units of solar radiation, also interpreted as most of the time there are no high readings of energy. This could be a highly noticeable remark to drop the information, as most information containing 0 as a value could not contribute to the model. However, the 0 reading could be explained by the recorded instances happening at night time or being a characteristic of Seoul. We can also expect the possibility of the model creating a relationship between this and other features, meaning that this data could identify possible groups in this highly skewed data and therefore is kept and will be fed to the model.

1.8 The histogram on rainfall shows most of the values fall near 0, below approximately 2.5. The distribution could be consider to not be a feature to study or even transformed

as most of data falls on a small range however we have outliers and a maximum record of 35.0 milimeters in rain, these outliers of high rain instances could provide still a high chance influence in predicting, meaning with small data it can be almost absolute assumption that high rainfall produces very low rental behaviour therefore is kept for the model.

1.9 The histogram on the snowfall follows a similar behaviour, with most of the data falling below 0.5 cm and, due to the same logic, it could be argued that this information can be dropped or transformed; however, the maximum value occurring at 5.1 cm also gives the possibility of a high indicator of rental behaviour. These outliers could play an important role in the prediction model; therefore, it's kept.

1.10 The histogram on holidays was produced to understand a visual representation of the ratio between non-holiday or workless days. These days are important as it highly impacts the routine of the citizens of Seoul. Important to mention that this data can be fed without transformation; this is a binary value, so there is no use in exploring deeper into the distribution and mean; therefore kept as is for the model. I do propose the consideration that it could be transformed or grouped with more information, or even dropped, to explain, in a deeper manner, the behaviour of holidays. Meaning that holidays could impact in a highly unpredictable way as is, since holidays could produce high rental behaviour or even a lower one due to the possibility on lower citizens being in Seoul during these instances.

B. Distribution with Boxplot

We now proceed to create box-and-whisker graphs of box and whiskers for all continuous variables; graphing for categorical and discrete variables is not worth doing. We are left with the next graphs. It will be useful to visualize possible outliers better that could not be well noticed on the histogram graphs.

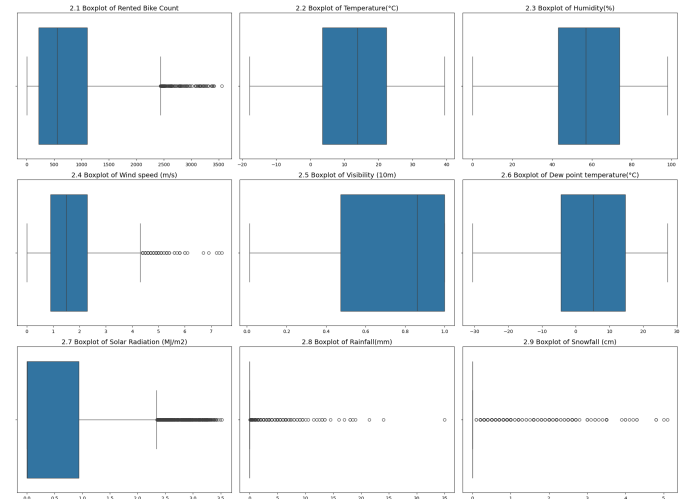


Fig. 2. Boxplot of continuous variables and holiday distribution.

2.1 To describe it, the box itself is located on the left side, and the median is around 500 counts on the left side

of the box. Regarding distribution, it has wide whiskers or spread, with high outliers. The worth-mentioning information to spot on the Boxplot for the Rented Bike Count is the high number of outliers. This could be an early remark that many real possible specific combinations of X features can provide high rental behaviour. of outliers.

2.2 , 2.3 , 2.5 , 2.6 After inspecting the boxplots produced for Temperature, Humidity percentage, Visibility and Dew point temperature, we can state that there are no noticeable outliers. Describing the graphs, it shows a highly packed box distribution in the data. The thing to consider is that the median for Temperature its notices it falls around 15 degrees and is located on the half of the box, making it a symmetrical distribution. For Humidity, the median falls around 55%, and it is more on the left side of the box. For Visibility, the length of the box is bigger but still skewed back, with the median located at around 0.85 on the right side of the box. Finally, for Dew point, the box is heavily packed and the median cuts symmetrically through the box, falling around 8 degrees. Even with large whiskers, the lack of outliers and a highly packed box is a sign of a more balanced or spread-out distribution. It indicates that the data is well scattered and most of the data has fallen within a statistical scope. It is important to take this factor in consideration when analysing further data also open to the possibility of falling by nature into a normal distribution, at this point the histograms 1.2, 1.3, 1.5, 1.6 didnt presented a strong characteristic of normality however it could be an early sign that this is due for the limitations on the data sample and that having a larger sample size of more than 1 year it could be appreciated a normal distribution then having solid new information on how to treat and transform this variables.

2.4 , 2.7 , 2.8 , 2.9 To describe Wind Speed, the box is located on the left side with a median around 1.5; the box almost looks symmetrical, but the median falls more to the left side. The Solar radiation Box has the median close to the left end of the box, indicating a high concentration of data around 0.0. Is worth mentioning that, in the case of Rainfall and Snowfall, the Box and whiskers are present in this graph; this is because it is a highly concentrated and skewed distribution, so both boxes and whiskers are small and located on the left side of their corresponding graph. Important to keep in mind when analysing further data. When checking the whiskers for Wind Speed. Solar Radiation, Rainfall, and Snowfall present a high number of outliers. As mentioned before, all of these points do not mean they are negative characteristics for the model, and they will not spoil it; they are real possible events that could happen in Seoul. The purpose is not to eliminate or impute new values to these outliers, but to understand how they could affect the results of the model and how to manage and interpret the data, as the mean and variance for these values will have another perspective.

IV. PRINCIPAL COMPONENT ANALYSIS (PCA)

V. MODEL CONSTRUCTION (MANUAL)

VI. MODEL CONSTRUCTION (FRAMEWORK)

VII. RESULTS AND ANALYSIS

VIII. VERSIONS AND COMPARISON

IX. CONCLUSION

REFERENCES

- [1]
- [2]